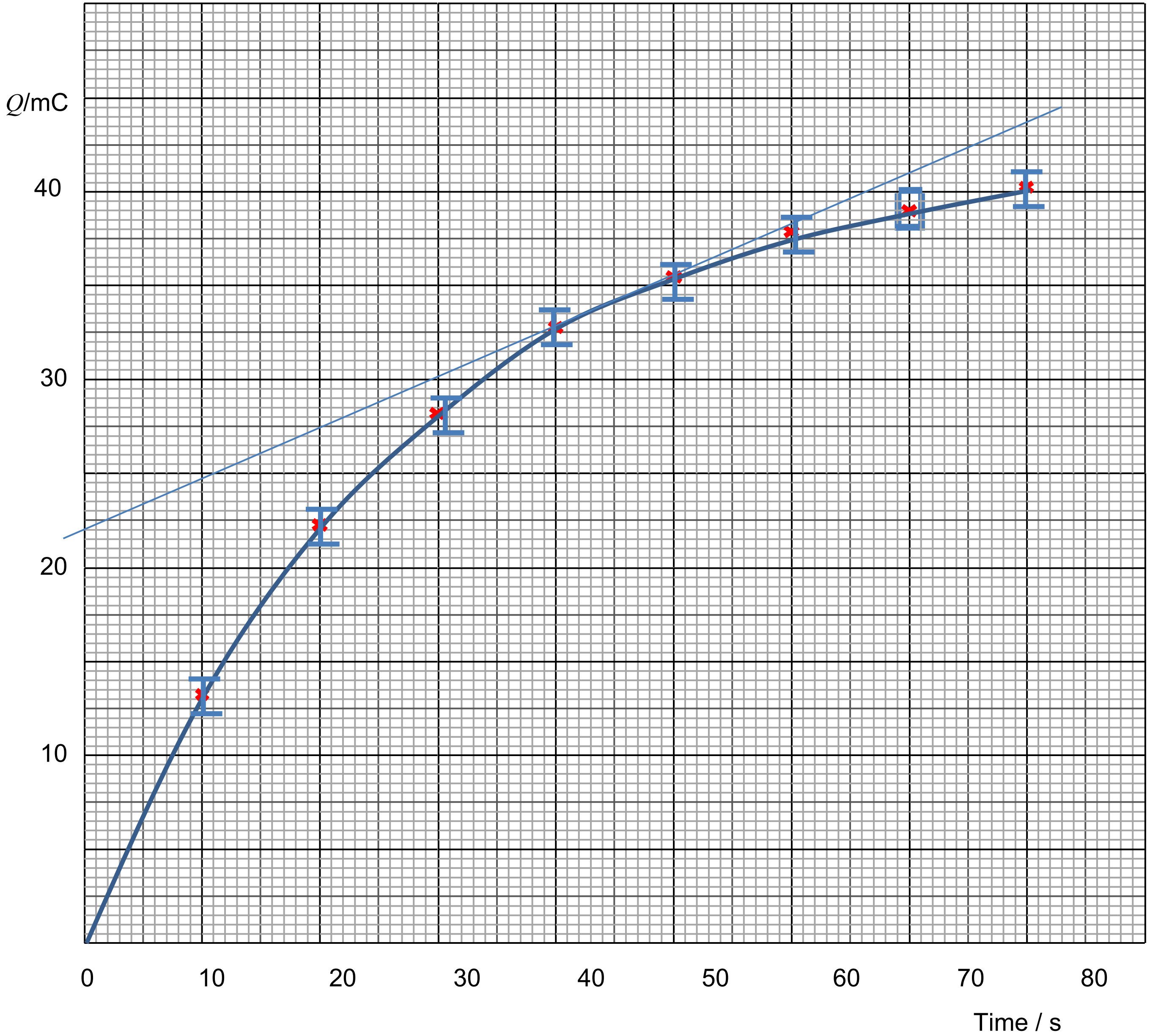
Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	*(a)*	(i)	Use of $C = \frac{\varepsilon_0 A}{d}$ [73.6 pF] [1] Use of $E = \frac{1}{2}CV^2$ **or** combination of $E = \frac{1}{2}QV$ **and** $Q = CV$ [1] Rearrangement $\left(\frac{2Ed}{\varepsilon_0 A}\right)$ **OR** substitution in both equations [1] Correct answer = 833 [V] [1]	1 1	1 1		4	3	
		(ii)	[Pd supplies/leads to] charges on plates (1) accept electric field set-up [This set-up] can do work / [electrostatic] PE stored / energy stored <u>in field</u> (1) accept can provide current after pd removed	2			2		
		(iii)	Any 3 ($\times$ 1) valid points e.g. • Repeat experiment / more tests / repeatability • Repeats by other research groups / industry / reproducibility • Tests for safety [of dielectric] • Tests for lifetime [of dielectric] • Cost effectiveness / availability • Environmental impact • Check for charge leakage			3	3		
	(b)	(i)	2.2 mF $\times$ 0.5 V seen **or** clear % usage shown		1		1		1
		(ii)	28.2 (1) 37.6 (1)		2		2		2
		(iii)	Numbers on y-axis i.e. 10, 20, 30, 40 (1) Both points plotted correctly **ecf** < small square (1) y-error bars correct < small square (1) Good curve of best fit **ecf** (2) If flawed but valid attempt at best fit e.g. hairy line, not smooth, missing one of the bars award 1 mark only		5		5		5

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(iv)	Obtaining Q_0 = 19.1 × 2.2 mF = 42 ± 2 m[C] (1) Using 63 % of fully charged **OR** substituting values into $Q = Q_0\left(1 - e^{-\frac{t}{RC}}\right)$ **OR** taking logs correctly (1) T **or** RC = 27 ± 2 [s] (no marks for 26.4 s) **OR** Q value correct after substituting (1)		3		3	3	3
		(v)	Good tangent drawn at 45 s **ecf** (1) Gradient = 0.27 m[A] **ecf** (1)		2		2		2
	(c)		**Any 5 × (1) from:** - Shape of graph is good - Line of best fit goes through error bar(s) - RC = 26.4 s - RC in good agreement with 27 s (**ecf** available) - $I_0 = \frac{19.1}{12\,000}$ = 1.59 m[A] - Gradient of graph decreases / gradient is the current - So current equation is in good agreement – only award mark if current decreasing implied - Substitution into either equation for confirmation			5	5	2	5
			Question 1 total	**4**	**15**	**8**	**27**	**8**	**18**


Q/mC
40
30
20
10
0
10
20
30
40
50
60
70
80
Time / s